UNIVERSITY OF UYO, UYO
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF MATHEMATICS
2022/2023 FIRST SEMESTER EXAMINATION
MTH 213: LINEAR ALGEBRA I

Time: 2hrs

Instruction: Answer **Question 1** and any other **three** questions

1(a) Give an exhaustive definition of a vector space?

(b) When is a vector space said to be;
(i) Finite dimensional
(ii) Infinite dimensional

(c) Show that the set of all polynomials is a vector space.

(d) What is a basis of a vector space?

2(a) Define each of the following terms along with their appropriate equations.
(i) Linear combination
(ii) Linear dependence
(ii) Linear independence

(b) Express the vector $u_1 = (5, 2, -1)$ as a linear combination of vectors $v_1 = (2, 3, 0)$, $v_2 = (1, 0, 2)$, and $v_3 = (4, 2, 1)$.

3(a) Suppose a linear operator $F: \mathbb{R}^2 \to \mathbb{R}^2$ is defined by;

$$F(x_1, x_2) = (2x_1 - 3x_2, 4x_1 + 5x_2)$$

Find the matrix representation of the linear operator F relative to the basis $S = \{u_1, u_2\}$, where $u_1 = (1, 3)$ and $u_2 = (2, 5)$.

(b) Find the matrix representation of the linear operator F in 3(a) above relative to the standard basis of $\mathbb{R}^2$.

4(a) Show that the mapping $G: \mathbb{R}^2 \to \mathbb{R}^2$ defined by;

$$G(x_1, x_2) = (x_1, x_1 + x_2)$$

is a linear map.

(b) Suppose the mapping G in 4(a) above is redefined as;

$$G^*(x_1, x_2) = (x_1 x_2, x_1)$$

using the vectors $u = (a, b)$ and $v = (a^*, b^*)$, show that the redefined mapping G^* is not linear.

5(a) Given the matrix $A = \begin{pmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 4 \end{pmatrix}$ and the column matrix $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

Write an expression for $X^T AX$.

(b) Investigate the linearity of the vectors $v_1 = 2i + 3j - 4k$, $v_2 = i - 2k$, and $v_3 = 4i - 2j + k$

6(a) Define a subspace of a vector space.

(b) Consider the following subspaces of $\mathbb{R}^4$ namely $X = \{(1, 2, 1, 1), (0, 1, 1, 1), (1, 0, 1, 1)\}$ and $Y = \{(1, -2, 2, 1), (1, -3, 4, 3), (-3, 2, 3, 4)\}$. Find

(i) dim $(X + Y)$
(ii) dim $(X \cap Y)$